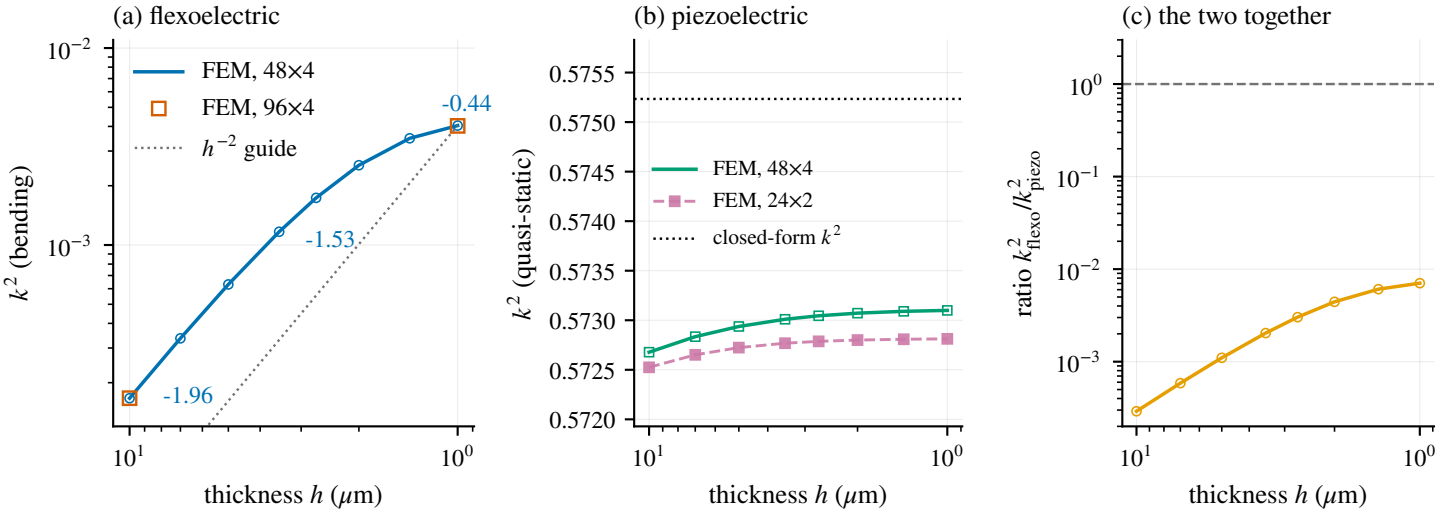
(a) flexoelectric
FEM, 48×4
FEM, 96×4
h^{-2} guide
-0.44
-1.53
-1.96
k^2 (bending)
thickness h (μm)
(b) piezoelectric
FEM, 48×4
FEM, 24×2
closed-form k^2
k^2 (quasi-static)
thickness h (μm)
(c) the two together
ratio $k^2_{\text{flexo}}/k^2_{\text{piezo}}$
thickness h (μm)